# Screening for Gravitationally Lensed Gravitational-Wave Events via Skymap Overlap

Kaitlyn Pak, Dr. Ignacio Magaña Hernandez
*Carnegie Mellon University, Department of Physics*

## Motivation

The 2017 detection of GW170817 — a binary neutron star merger at ~40 Mpc — marked the dawn of multimessenger gravitational-wave astronomy. The gravitational-wave signal was followed 1.7 seconds later by the short gamma-ray burst GRB 170817A, and ~11 hours later by the optical kilonova AT 2017gfo in the galaxy NGC 4993, confirming that binary neutron star mergers produce short GRBs and heavy elements. Follow-up was possible only because the GW sky localization was tight enough (~30 deg²) to narrow the search to a tractable list of host-galaxy candidates. Most events are localized far less precisely. This project develops a framework for quantifying skymap overlap across events and evaluating whether a given overlap is unusual — a first-pass screening tool for candidate lensed pairs and multimessenger follow-up.

## Methods

Gravitational-wave events and their best-available skymaps are retrieved from GraceDB. A skymap-overlap statistic (Figure 0.5) is computed in parallel across all 1.35 million unique event pairs to quantify how strongly their localization probability distributions coincide. Under the lensing hypothesis — that two events share a source — the joint sky localization is given by the normalized product (Figure 0.6), which tightens dramatically when the events agree on sky position. The resulting distribution (Figure 1) identifies unusually high-overlap pairs and estimates how unusual each overlap is relative to the empirical distribution. Low overlap indicates disjoint sky regions; high overlap, coincident ones.

## Results

Across 1.35 million BBH event pairs, the overlap distribution peaks near $\mathcal{O} \approx 1\ \mathrm{sr}^{-1}$, the baseline for finite-area localizations on the sphere. Twenty-five pairs exceed $\mathcal{O} = 100$. The highest overlap ($\mathcal{O} \approx 564$ for S250611f & S250611g) corresponds to two pipeline detections of the same trigger, validating the metric. Subsequent ranked pairs are separated by months, with the top time-separated candidate (S250727dc & S250122c, 186 days apart, $\mathcal{O} \approx 324$) showing a joint 90% region 44% the size of the tighter individual region.

$$\mathcal{O} = \int P_1(\Omega)\, P_2(\Omega)\, d\Omega$$

Figure 0.5

$$P_{\text{joint}}(\Omega) = \frac{P_1(\Omega)\, P_2(\Omega)}{\int P_1(\Omega')\, P_2(\Omega')\, d\Omega'}$$

Figure 0.6

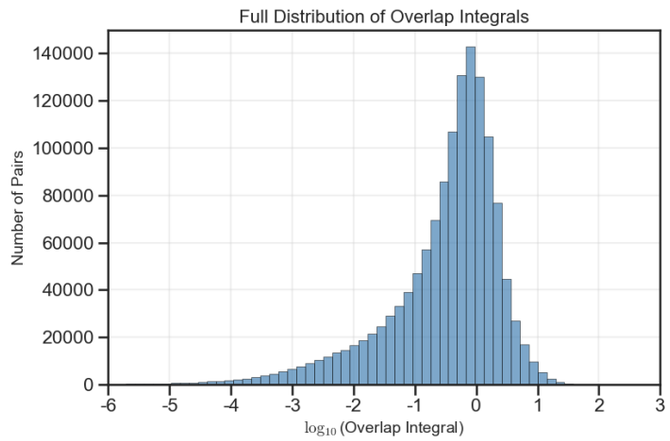


Figure 1

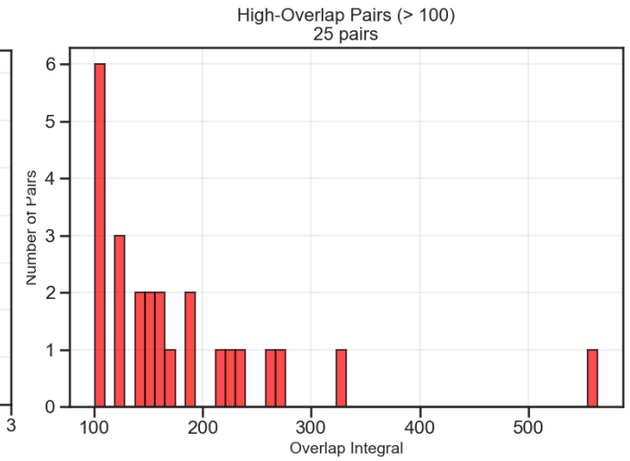


Figure 2

*Figure 0.5* shows the overlap integral used on each event comparison. *Figure 0.6* shows the joint sky normalized product formula. *Figure 1* shows a histogram of every overlap integral calculated. *Figure 2* shows a closer view of the highest overlap pairs.

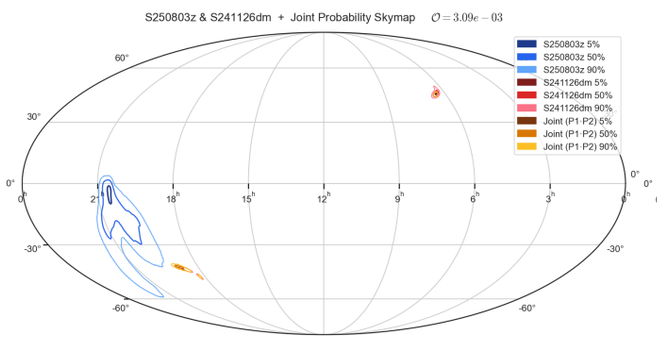


Figure 3

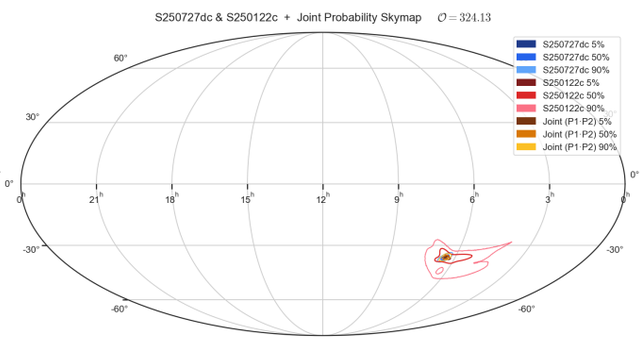


Figure 4

*Figure 3* shows two GW events with one of the lowest overlap (5th percentile). Conversely, *Figure 4* shows the two distinct GW events with the highest overlap.

**Top 10 highest-overlap pairs**

| Rank | Event 1 | Event 2 | Days apart | Overlap |
|---|---|---|---|---|
| 1 | S250611f | S250611g | 0 | 563.905 |
| 2 | S250727dc | S250122c | 186 | 324.130 |
| 3 | S250611f | S250611h | 0 | 269.681 |
| 4 | S250712bt | S250108x | 185 | 261.706 |
| 5 | S241202bi | S231204ak | 364 | 234.844 |
| 6 | S241231bg | S240919bn | 103 | 225.325 |
| 7 | S250611g | S250611h | 0 | 221.016 |
| 8 | S241218at | S240616cm | 185 | 191.556 |
| 9 | S240925n | S250227e | 155 | 184.038 |
| 10 | S250611f | S250611e | 0 | 167.459 |

*Figure 5.* Table of top 10 highest-overlap pairs.

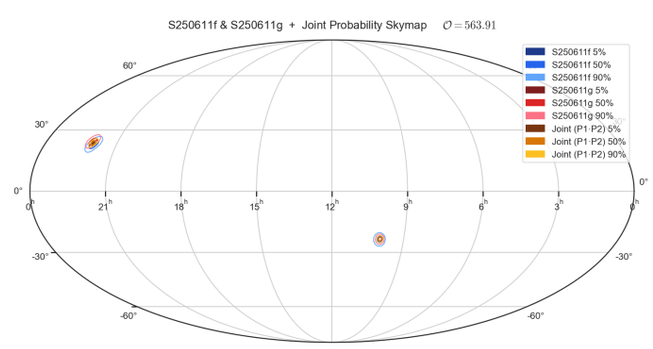


*Figure 6.* Joint skymap for S250611f & S250611g ($\mathcal{O} = 563.91$); same-day duplication detection

## Future Applications

While this framework already serves as a first-pass screen for candidate lensed pairs, the lensing application can be developed much further. A principled null model — for example, simulating randomized sky positions rather than relying on the empirical pair distribution — would yield well-defined p-values for overlap significance. Beyond sky position, true lensed pairs are expected to agree in detector-frame masses, spins, and waveform morphology; incorporating these constraints would sharpen candidate selection. Highest-overlap pairs should also be cross-checked against electromagnetic transient catalogs to test whether unusual sky coincidences extend to the multimessenger domain. Ultimately, this pipeline can be released as an open-source package, enabling the broader gravitational-wave community to apply skymap-overlap screening to future observing runs.

## Conclusion

This project develops a statistical framework for comparing gravitational-wave skymaps and evaluating whether a given overlap is unusual. Most of the 1.35 million event pairs analyzed show low overlap, while a small subset stand out — including candidates worth investigating under the gravitational-lensing hypothesis, where two detections may share a single source. As a screening tool, this approach drastically reduces the follow-up burden by flagging the rare high-overlap pairs deserving deeper analysis and ruling out the vast majority where the events are clearly localized to different parts of the sky. Skymap overlap alone is not enough to prove a shared origin, but it provides a useful first-pass method for prioritizing candidate lensed pairs.

## Acknowledgments

*Data: LIGO–Virgo–KAGRA O4 from GraceDB. Funded by the Porges Family Fund for Undergraduate Research. Thanks to Dr. Ignacio Magaña Hernandez for advising.*

2025-2026 PORGES FELLOW
This project was made possible by the generosity of the Porges Family Fund for Undergraduate Research